Torsional friction in planar contact

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Abstract

Tactile sensors are widely used to detect slip, but much less is known about what can actually be inferred about frictional contact limits from tactile measurements. This question matters because detecting that slip is occurring or imminent is not the same as knowing which tangential and torsional loads the contact can still support. That distinction is important for control design. In this report, we study this question for a single planar contact patch under quasi-static Coulomb friction. We first show that coarse normal force information, consisting of the total normal force and center of pressure, is generally insufficient to determine the admissible tangential wrench set: two contacts can have the same total normal force and CoP while supporting different combinations of tangential force and torsional moment. We then study what can still be inferred from partial measurements. In particular, we derive upper and lower bounds on the admissible wrench set from coarse measurements and from spatially distributed taxel-wise normal forces. The latter provide tighter bounds by reducing the ambiguity on how the normal force is distributed over the contact patch.

1 Introduction

Modern tactile sensors provide rich local information about contact pressure, shape, and deformation. Yet tactile features used for control are often selected mainly from empirical performance, for example contact pose, resultant force, image features, or local pressure measurements. A more basic question is less often made explicit: given a particular tactile measurement, what can actually be inferred about the physical interaction at the contact? We study this question in the setting of a rigid planar contact patch under Coulomb friction. Rather than trying to characterize every possible tactile signal or manipulation task, we focus on one concrete control-relevant quantity: the set of tangential forces and torsional moments that the contact can support without violating the friction model. The key difficulty is that this admissible wrench set depends on how the normal load is distributed over the patch. Coarse measurements such as the total normal force and center of pressure constrain this load distribution, but do not determine it uniquely. Spatial tactile measurements provide additional information and therefore reduce this ambiguity. The objective of this report is to make this relationship explicit. We ask **how the admissible frictional wrench set can be bounded from different levels of normal force information**, from total force and CoP to spatially distributed taxel-wise forces, **without requiring the full normal pressure field**.

2 Preliminaries (measures)

We briefly introduce the measure-theoretic notation used in the following. Only the notions needed for the contact model are recalled here; standard reference is [1].

2.1 Force as a measure.

Let $P \subset \mathbb{R}^2$ denote the contact patch, equipped with its Borel σ-algebra $\mathcal{B}(P)$. A *nonnegative measure* $\nu_N : \mathcal{B}(P) \to [0, +\infty]$ assigns to every measurable subset $A \subseteq P$ a nonnegative scalar $\nu_N(A)$, and is countably additive, i.e. for any pairwise disjoint measurable sets $A_i \subseteq P$,

$$\nu_N\left(\bigcup_i A_i\right) = \sum_i \nu_N(A_i). \tag{1}$$

In the present context, ν_N represents the (spatially distributed) normal force, i.e. $\nu_N(A)$ is the normal force carried by the region A. Hence $\nu_N(A)$ has units of force (N), and the resultant normal force is $f_N = \nu_N(P)$. We restrict attention to *finite nonnegative Radon measures*, denoted $\nu_N \in \mathcal{M}_+(P)$. Informally, a Radon measure is a "well-behaved" measure. In particular, on the compact sets considered here, it is finite and its value on measurable sets can be approximated from compact subsets. For our purpose, it essentially means that our measure does not assign an infinite force to P and it can represent either the force induced by an ordinary pressure field or an idealized concentrated point/line load.

We represent the tangential force by an $\mathbb{R}^2$-valued finite Radon measure $\nu_T = (\nu_{T,1}, \nu_{T,2})$ where each component $\nu_{T,i}$ is a finite signed Radon measure on P. Thus, for every measurable region $E \subseteq P$, the resultant tangential force carried by E is

$$\nu_T(E) = \begin{bmatrix} \nu_{T,1}(E) \\ \nu_{T,2}(E) \end{bmatrix} \in \mathbb{R}^2, \tag{2}$$

2.2 Area measure and pressure fields.

Let dA denote the usual area element on the planar patch P. A *normal pressure field* is a nonnegative function $\rho_N : P \to \mathbb{R}_+$ with units of force per unit area. If $\rho_N \in L^1_+(P)$, which simply means that ρ_N has finite integral over P

$$\rho_N(x) \geq 0 \quad \text{a.e.}, \qquad \int_P \rho_N(x)\, dA < \infty. \tag{3}$$

then it defines a finite normal force measure: the force carried by any region $E \subseteq P$ is given by

$$\nu_N(E) = \int_E \rho_N(x)\, dA. \tag{4}$$

which we also write equivalently as $d\nu_N(x) = \rho_N(x)\, dA$. In that case, ρ_N is called the *density* of the normal force measure ν_N with respect to area[1]. Not every normal force measure admits such a pressure field. For example the Dirac measure $\nu_N = f_N \delta_c$ represents an idealized normal force f_N concentrated at the single point c. It cannot be represented by an ordinary pressure field over area. In this note we adopt the measure-theoretic language (instead of only L^1_+ densities) so that we cover both "classical" pressure fields and idealized concentrated loads.

This notation allows ordinary pressure fields and concentrated loads to be handled by the same integration equations. For a measurable scalar or vector function g, the integral

$$\int_P g(x)\, d\nu_N(x) \tag{5}$$

can be interpreted as a force-weighted spatial integral. Whenever ν_N admits a normal pressure field ρ_N,

$$\int_P g(x)\, d\nu_N(x) = \int_P g(x)\rho_N(x)\, dA. \tag{6}$$

[1] More generally, the Radon-Nikodym theorem guarantees the existence of a density for measures under suitable assumptions, such as "absolute continuity" which means physically that a subset of zero area (e.g. a point or a line) carries zero normal force. See [1], Theorem 1.30, p. 50

For a Dirac measure,

$$\int_P g(x)\, d(f_N \delta_c)(x) = f_N g(c). \tag{7}$$

2.3 Moments of the normal force measure.

The zeroth moment of ν_N is its total measure,

$$m_0(\nu_N) \triangleq \int_P 1\, d\nu_N = \nu_N(P) = f_N. \tag{8}$$

In the present mechanical interpretation, this is the *resultant normal force*. The first moment is

$$m_1(\nu_N) \triangleq \int_P x\, d\nu_N(x) \in \mathbb{R}^2. \tag{9}$$

If $f_N > 0$, the *center of pressure* is the normalized first moment,

$$c[\nu_N] \triangleq \frac{m_1(\nu_N)}{m_0(\nu_N)} \tag{10}$$

$$= \frac{1}{f_N} \int_P x\, d\nu_N(x). \tag{11}$$

Hence knowing the total normal force and the center of pressure is equivalent to knowing the zeroth and first moments of the normal force measure $m_0(\nu_N) = f_N, m_1(\nu_N) = f_N c[\nu_N]$.

Barycenters and local CoPs. More generally, the *barycenter* of a nonnegative measure ν carried by a set E with $\nu(E) > 0$ is its normalized first moment $\frac{1}{\nu(E)} \int_E x\, d\nu(x)$. For the normal force measure on P, the barycenter is the CoP, $c[\nu_N]$. For the restriction of ν_N to a subset $E \subseteq P$, such as a taxel, it is the *local CoP* of the normal force carried by E.

3 Problem formulation

3.1 Planar contact model

Assume that the planar contact patch $P \subset \mathbb{R}^2$ is compact and convex, and let $\nu_N \in \mathcal{M}_+(P)$ and ν_T be the normal and tangential force measures introduced above. We assume

$$f_N = m_0(\nu_N) > 0, \tag{12}$$

so that the center of pressure c is well-defined. Since P is convex and ν_N is nonnegative, $c \in P$. We further assume a known and spatially uniform Coulomb friction coefficient $\mu > 0$. We impose Coulomb friction in measure form by requiring that there exists a measurable density $\gamma_T : P \to \mathbb{R}^2$ of ν_T with respect to ν_N such that

$$d\nu_T(x) = \gamma_T(x)\, d\nu_N(x), \qquad \|\gamma_T(x)\|_2 \leq \mu \quad \nu_N\text{-a.e. on } P. \tag{13}$$

We call γ_T the *friction ratio field*. It can be interpreted as the tangential force per unit normal force. The above condition expresses both that tangential friction force can only be carried where normal force is present and that its local magnitude is bounded by μ[2]. When ν_N admits the normal pressure field ρ_N introduced above, i.e. $d\nu_N = \rho_N\, dA$, the tangential force measure also admits a density with respect to area,

$$\rho_T(x) = \gamma_T(x)\rho_N(x), \tag{14}$$

[2]In particular, $d\nu_T = \gamma_T\, d\nu_N$ implies $\nu_T \ll \nu_N$.

called the *tangential traction field*. The measure-form Coulomb condition then reduces to the classical local condition

$$\|\rho_T(x)\|_2 \le \mu \rho_N(x) \qquad \text{a.e. on } P. \tag{15}$$

We define the *resultant tangential force* and *resultant torsional moment* about the center of pressure c as

$$f_T \triangleq \nu_T(P) = \int_P \gamma_T(x)\, d\nu_N(x), \qquad \tau_N \triangleq \int_P (x-c) \times \gamma_T(x)\, d\nu_N(x). \tag{16}$$

We finally define the skew-symmetric matrix

$$J \triangleq \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \tag{17}$$

so that for $x, y \in \mathbb{R}^2$ $x \times y = (Jx)^\top y$

Definition 3.1 (Admissible tangential wrench set)**.** *Define a normal force measure $\nu_N \in \mathcal{M}_+(P)$. We define admissible tangential wrench set :*

$$\mathcal{W}(\nu_N) \triangleq \left\{ (f_T, \tau_N) \in \mathbb{R}^2 \times \mathbb{R} : \begin{array}{l} \exists\, \gamma_T : P \to \mathbb{R}^2 \text{ measurable such that} \\ \|\gamma_T(x)\|_2 \le \mu \quad \nu_N\text{-a.e.}, \\ f_T = \displaystyle\int_P \gamma_T(x)\, d\nu_N(x), \\ \tau_N = \displaystyle\int_P (x - c[\nu_N]) \times \gamma_T(x)\, d\nu_N(x) \end{array} \right\}. \tag{18}$$

The set $\mathcal{W}(\nu_N)$ is a non-empty compact subset of $\mathbb{R}^3$ containing all resultant tangential wrenches (tangential force and torsional moment) that are compatible with the normal force measure ν_N and the local Coulomb friction condition (13).

Proposition 3.2 (Support function of the admissible tangential wrench set)**.** *For any $\nu_N \in \mathcal{M}_+(P)$ and for every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, the support function of $\mathcal{W}(\nu_N)$ is*

$$h_{\mathcal{W}(\nu_N)}(q) = \mu \int_P \|a + bJ(x - c[\nu_N])\|_2 \; d\nu_N(x). \tag{19}$$

Proof. By definition, $\forall q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$

$$h_{\mathcal{W}(\nu_N)}(q) = \sup_{(f_T, \tau_N) \in \mathcal{W}(\nu_N)} \left(a^\top f_T + b\tau_N \right). \tag{20}$$

For any friction ratio field γ_T

$$a^\top f_T + b\tau_N = \int_P \left[a + bJ(x - c[\nu_N]) \right]^\top \gamma_T(x)\, d\nu_N(x). \tag{21}$$

Using Cauchy–Schwarz and $\|\gamma_T(x)\|_2 \le \mu$,

$$a^\top f_T + b\tau_N \le \mu \int_P \|a + bJ(x - c[\nu_N])\|_2 \; d\nu_N(x). \tag{22}$$

The bound is attained by choosing $\gamma_T(x)$ with magnitude μ and aligned with $a + bJ(x - c[\nu_N])$ whenever this vector is nonzero:

$$\gamma_T^\star(x) = \begin{cases} \mu \frac{a + bJ(x - c[\nu_N])}{\|a + bJ(x - c[\nu_N])\|_2} & \text{if } \|a + bJ(x - c[\nu_N])\|_2 \neq 0 \\ 0 & \text{otherwise} \end{cases} \tag{23}$$

□

Symbol	Meaning	Remark	Lives in
ν_N	normal force measure	true $\nu_N^\star$ is unknown	$\mathcal{M}_+(P)$
$c[\nu_N]$	CoP induced by measure ν_N	depends on ν_N	$P \subset \mathbb{R}^2$
$q = (a, b)$	wrench support direction	prescribed	$\mathbb{R}^2 \times \mathbb{R}$
$\mathcal{M}$	measurement operator	fixed by the sensing model	$\mathcal{M}_+(P) \to \mathbb{R}^r$
y	measurement value	observed data from $(\mathcal{M}, \nu_N)$	$\mathbb{R}^r$
$\Pi_{\mathcal{M}}(y)$	all measures compatible with y	depends on $(\mathcal{M}, y)$	$\subseteq \mathcal{M}_+(P)$
$\mathcal{C}_{\mathcal{M}}(y)$	CoPs compatible with y	depends on $(\mathcal{M}, y)$	$\subseteq P$
$\underline{h}_{\mathcal{M}}(q; y)$	measurement-conditioned lower bound	depends on $(\mathcal{M}, y, q)$	$\mathbb{R}$
$\overline{h}_{\mathcal{M}}(q; y)$	measurement-conditioned upper bound	depends on $(\mathcal{M}, y, q)$	$\mathbb{R}$

Table 1: Notations.

3.2 Measurement-conditioned support bounds

We focus here on the case where we do not know the *true* force measure $\nu_N^\star$ but we have only access to some measurements of it (e.g. resultant normal force, CoP, etc.). The goal is to investigate what can be known about the support of $\mathcal{W}(\nu_N^\star)$ given partial measurements of $\nu_N^\star$.

Definition 3.3 (Measurement operator)**.** *We define a measurement operator as a function $\mathcal{M} : \mathcal{M}_+(P) \to \mathbb{R}^r$. We assume that its components are linear integral measurements of the form*

$$\mathcal{M}(\nu_N) = \begin{bmatrix} \int_P \phi_1(x)\, d\nu_N(x) \\ \vdots \\ \int_P \phi_r(x)\, d\nu_N(x) \end{bmatrix}, \qquad \forall \nu_N \in \mathcal{M}_+(P) \tag{24}$$

where the prescribed measurable functions $\phi_j : P \to \mathbb{R}$ are called the measurement functions. In the following, we denote a measurement value (or observation) induced by the (unknown) true force measure $\nu_N^\star$ as $y = \mathcal{M}(\nu_N^\star) \in \mathbb{R}^r$.

Definition 3.4 (Measurement-compatible normal force family)**.** *Given a measurement operator $\mathcal{M} : \mathcal{M}_+(P) \to \mathbb{R}^r$ and the observation $y \in \mathbb{R}^r$, we define the set of normal force measures that are compatible with observation y*

$$\Pi_{\mathcal{M}}(y) \triangleq \{\nu_N \in \mathcal{M}_+(P) : \mathcal{M}(\nu_N) = y\}. \tag{25}$$

The set $\Pi_{\mathcal{M}}(y)$ contains all the normal force measures that are indistinguishable under measurement y. Note that this is the pre-image of y through $\mathcal{M}$: $\Pi_{\mathcal{M}(y)} = \mathcal{M}^{-1}[\{y\}]$.

Assumption 3.5. *From now on, we assume $\forall \nu_N \in \Pi_{\mathcal{M}}(y)$, $\nu_N(P) > 0$. Purpose: we restrict the measurement-compatible family to measures that have a well-defined CoP.*

Definition 3.6 (Set of measurement-compatible CoPs)**.** *We define*

$$\mathcal{C}_{\mathcal{M}}(y) \triangleq \{c[\nu_N] : \nu_N \in \Pi_{\mathcal{M}}(y)\} \tag{26}$$

as the set of CoPs compatible with the observation $y \in \mathbb{R}^r$.

Given an observation y of the unknown true force measure $\nu_N^\star$, we can compute measurement-conditioned lower/upper bounds on the support function $h_{\mathcal{W}(\nu_N^\star)}$ by optimizing over $\Pi_{\mathcal{M}}(y)$.

Definition 3.7 (Measurement-conditioned support bounds)**.** *Let $\mathcal{M} : \mathcal{M}_+(P) \to \mathbb{R}^r$ be a measurement operator as defined above and let $y = \mathcal{M}(\nu_N^\star)$ denote the observation induced by the*

unknown true normal force measure $\nu_N^\star$. For a wrench direction $q \in \mathbb{R}^2 \times \mathbb{R}$, we define the measurement-conditioned bounds in direction q:

$$\underline{h}_{\mathcal{M}}(q;y) \triangleq \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{W}(\nu_N)}(q), \tag{27}$$

$$\overline{h}_{\mathcal{M}}(q;y) \triangleq \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{W}(\nu_N)}(q). \tag{28}$$

Remark 3.8 (How to use the bounds $\underline{h}_{\mathcal{M}}(q;y), \overline{h}_{\mathcal{M}}(q;y)$)**.** *A practical consequence of knowing the bounds is that we can test the slip feasibility of a candidate tangential wrench w: guaranteed feasible if $\forall q,\ \langle q, w\rangle \le \underline{h}_{\mathcal{M}}(q;y)$. Conversely, if $\exists q,\ \langle q, w\rangle > \overline{h}_{\mathcal{M}}(q;y)$, then w is infeasible for every normal force measure compatible with the measurement y*[3]*.*

Remark 3.9 (Relation with inner/outer wrench sets)**.** *We can relate the bounds of Definition 3.7 to outer/inner approximations of the admissible wrench set*

$$\mathcal{W}^+(y) \triangleq \bigcup_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{W}(\nu_N), \qquad \mathcal{W}^-(y) \triangleq \bigcap_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{W}(\nu_N). \tag{29}$$

Then $h_{\mathcal{W}^+(y)}(q) = \overline{h}_{\mathcal{M}}(q;y)$ and $h_{\mathcal{W}^-(y)}(q) \le \underline{h}_{\mathcal{M}}(q;y)$ (equality need not hold for the inner set). Nevertheless, the lower bound characterizes the inner set exactly through

$$\mathcal{W}^-(y) = \{w :\ \langle q, w\rangle \le \underline{h}_{\mathcal{M}}(q;y) \quad \forall q\}\,. \tag{30}$$

Computing the bounds of Definition 3.7 is difficult in general as it leads to a nonlinear optimization problem over measurement-compatible measures:

$$\underline{h}_{\mathcal{M}}(q;y) = \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mu \int_P \|a + bJ\,(x - c[\nu_N])\|_2 \, d\nu_N(x), \tag{31}$$

$$\overline{h}_{\mathcal{M}}(q;y) = \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mu \int_P \|a + bJ\,(x - c[\nu_N])\|_2 \, d\nu_N(x). \tag{32}$$

The objective is nonlinear in ν_N since the integrand depends on it through $c[\nu_N]$. Hence, this is not a standard Generalized Moment Problem (GMP) [2]. The next subsection collects the generic properties of (31)–(32) that are used for all the measurement operators studied afterwards.

3.3 General structure of the bounds

For $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$ and a reference point $c \in \mathbb{R}^2$, we write

$$g(x; q, c) \triangleq \|a + bJ(x - c)\|_2\,, \tag{33}$$

so that, by Proposition 3.2, $h_{\mathcal{W}(\nu_N)}(q) = \mu \int_P g(x; q, c[\nu_N])\, d\nu_N(x)$.

Lemma 3.10 (Properties of the integrand)**.** *For every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$ and $c, c' \in \mathbb{R}^2$:*

(i) $x \mapsto g(x; q, c)$ is convex on $\mathbb{R}^2$;

(ii) for all $x, x' \in \mathbb{R}^2$, $|g(x; q, c) - g(x'; q, c')| \le |b|\,(\|x - x'\|_2 + \|c - c'\|_2)$;

(iii) for every $\nu_N \in \mathcal{M}_+(P)$ with $\nu_N(P) > 0$, $\int_P g(x; q, c)\, d\nu_N(x) \ge \nu_N(P)\, g(c[\nu_N]; q, c)$. In particular, taking $c = c[\nu_N]$,

$$h_{\mathcal{W}(\nu_N)}(q) \ge \mu\, \nu_N(P)\, \|a\|_2. \tag{34}$$

[3] Because all support functions are positively homogeneous in q, it is sufficient in practice to search over one normalized representative of each nonzero ray. After fixing a consistent scaling between force and torque coordinates, one may for instance impose $\|q\|_2 = 1$.

Proof. (i) $g(\cdot; q, c)$ is a norm composed with an affine map. (ii) By the reverse triangle inequality and $\|J\|_2 = 1$, $|g(x; q, c) - g(x'; q, c')| \le |b|\,\|J((x-c)-(x'-c'))\|_2 \le |b|(\|x-x'\|_2 + \|c-c'\|_2)$. (iii) Apply Jensen's inequality: Let $m = \nu_N(P) > 0$. The mean of the measure $\pi = \nu_N/m$ is the CoP: $\int_P x\, d\pi(x) = \frac{1}{m}\int_P x\, d\nu_N(x) = c[\nu_N]$. Jensen's inequality thus reads

$$g(c[\nu_N]; q, c) \le \frac{1}{m}\int_P g(x; q, c)\, d\nu_N(x),$$

and multiplying by $m = \nu_N(P)$ gives the first claim. The particular case also follows directly, without Jensen's inequality (compare Remark 4.2 of Yann's draft): the uniform friction ratio field $\gamma_T \equiv \mu u$, with $u = a/\|a\|_2$ (any unit vector if $a = 0$), gives $f_T = \mu\,\nu_N(P)\,u$ and $\tau_N = \mu\left(J\int_P (x - c[\nu_N])\, d\nu_N(x)\right)^\top u = 0$, since the CoP is the barycenter of ν_N. Hence $(f_T, \tau_N) \in \mathcal{W}(\nu_N)$, and $h_{\mathcal{W}(\nu_N)}(q) \ge a^\top f_T + b\tau_N = \mu\,\nu_N(P)\|a\|_2$, which is (34). □

Lemma 3.11 (CoP slicing)**.** *For $c \in \mathbb{R}^2$, let $\Pi_{\mathcal{M}}(y; c) \triangleq \{\nu_N \in \Pi_{\mathcal{M}}(y) : \int_P (x-c)\, d\nu_N(x) = 0\}$ be the compatible measures whose CoP is c. Then $\Pi_{\mathcal{M}}(y) = \bigcup_{c\in\mathcal{C}_{\mathcal{M}}(y)} \Pi_{\mathcal{M}}(y; c)$, and*

$$\underline{h}_{\mathcal{M}}(q; y) = \inf_{c\in\mathcal{C}_{\mathcal{M}}(y)} \inf_{\nu_N\in\Pi_{\mathcal{M}}(y;c)} \mu\int_P g(x; q, c)\, d\nu_N(x), \tag{35}$$

$$\overline{h}_{\mathcal{M}}(q; y) = \sup_{c\in\mathcal{C}_{\mathcal{M}}(y)} \sup_{\nu_N\in\Pi_{\mathcal{M}}(y;c)} \mu\int_P g(x; q, c)\, d\nu_N(x). \tag{36}$$

For a fixed c, the inner problems are linear in ν_N under linear moment constraints, i.e. GMPs. In particular, if $\mathcal{C}_{\mathcal{M}}(y) = \{c\}$, the outer optimization disappears and bounds are GMPs.

Proof. Under Assumption 3.5, $\nu_N \in \Pi_{\mathcal{M}}(y)$ belongs to $\Pi_{\mathcal{M}}(y; c)$ if and only if $c = c[\nu_N]$, which gives the decomposition. On $\Pi_{\mathcal{M}}(y; c)$ the integrand of (31)–(32) is $g(x; q, c)$. □

Lemma 3.12 (Monotonicity with respect to information)**.** *Let $(\mathcal{M}, y)$ and $(\mathcal{M}', y')$ be two measurement operators and observations such that $\Pi_{\mathcal{M}'}(y') \subseteq \Pi_{\mathcal{M}}(y)$. Then $\mathcal{C}_{\mathcal{M}'}(y') \subseteq \mathcal{C}_{\mathcal{M}}(y)$ and, for every $q \in \mathbb{R}^2 \times \mathbb{R}$ and every $\nu_N \in \Pi_{\mathcal{M}'}(y')$,*

$$\underline{h}_{\mathcal{M}}(q; y) \le \underline{h}_{\mathcal{M}'}(q; y') \le h_{\mathcal{W}(\nu_N)}(q) \le \overline{h}_{\mathcal{M}'}(q; y') \le \overline{h}_{\mathcal{M}}(q; y). \tag{37}$$

This applies in particular to $\nu_N = \nu_N^\star$ when $y' = \mathcal{M}'(\nu_N^\star)$.

Proof. An infimum (resp. supremum) over a smaller set is larger (resp. smaller). □

Remark 3.13 (To clarify with Yann)**.** *The bounds (31)–(32) are defined as an infimum and a supremum. When they are attained is not fully clear to me yet. Points to clarify:*

(a) Are there generic conditions under which bounds are provably attained (and does it actually matter for our results) or do we need to prove it separately in each case ?

(b) For taxel measurements, $\phi_i = \mathbf{1}_{P_i}$ is discontinuous. My intuition is that extremal measures tend to put mass on taxel boundaries, which may belong to neighboring taxels. So I make the assumption that the measures are defined on the taxel closure (with boundary). Is this right? cf. Remark 5.7

Table 2 summarizes the measurement operators studied in the remainder of the report and the structure of the resulting problems, and Figure 1 illustrates what each of them observes. We next specialize (31)–(32) by increasing difficulty: the CoP is first measured, with coarse (Section 4) and then taxel-wise (Section 6) force information. It is then unknown (Section 7). The tightening of the bounds as information is added is studied in Section 8.

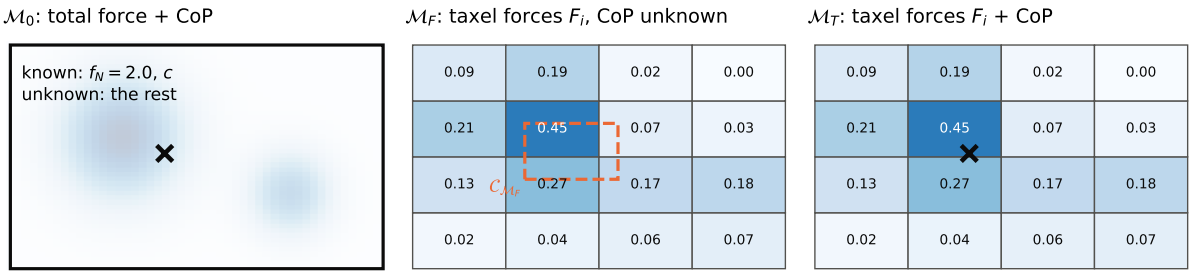


Figure 1: What each measurement operator observes for the same contact (4×4 taxels). $\mathcal{M}_0$: only f_N and the CoP, the pressure distribution (faded) is unknown. $\mathcal{M}_F$: the taxel forces F_i (cells filled with the mean pressure $F_i/|P_i|$), the CoP is only known to lie in $\mathcal{C}_{\mathcal{M}_F}(y_F)$ (dashed, Proposition 5.6). $\mathcal{M}_T$: taxel forces and CoP.

Measurement	Observed data	Compatible CoP set	Lower bound	Upper bound
$\mathcal{M}_0$ (Sec. 4)	$y_0 = (f_N, f_N c)$	$\{c\}$	closed form	LP (polygonal P)
$\mathcal{M}_T$ (Sec. 6)	$y_T = (y_0, y_F)$	$\{c\}$	convex (SOCP)	LP (polygonal taxels)
$\mathcal{M}_F$ (Sec. 7)	$y_F = (F_1, \dots, F_r)$	$\sum_{i=1}^{r} \frac{F_i}{f_N} P_i$	convex (SOCP)	nonconvex: $\max_c$ of $\mathcal{M}_T$ LPs

Table 2: Measurement operators, compatible CoP sets and structure of the resulting bounds.

4 $\mathcal{M}_0$ case (coarse measurement)

We first assume that the available measurements consist only of the resultant normal force f_N and the CoP c. **Known:** P, μ, f_N and c. **Unknown:** how $\nu_N^\star$ is distributed over P. Since the CoP is measured, Lemma 3.11 shows that both bounds are GMPs. We compute them explicitly and show that this partial information is generally not enough to determine the admissible tangential wrench set.

Definition 4.1 (Coarse normal force measurement)**.** *Given the true (unknown) normal force measure $\nu_N^\star \in \mathcal{M}_+(P)$, we define the coarse measurement as its zeroth and first moments*

$$y_0 = \mathcal{M}_0(\nu_N^\star) \triangleq \left(\int_P d\nu_N^\star(x),\ \int_P x\, d\nu_N^\star(x) \right) = (f_N, f_N c[\nu_N^\star]) \tag{38}$$

Remark 4.2. *The coarse measurement $y_0 = \mathcal{M}_0(\nu_N^\star)$ is a particular case of Definition 3.3, that extracts the zeroth and first moments through measurement functions:*

$$\phi_1(x) = 1, \qquad \phi_2(x) = x_1, \qquad \phi_3(x) = x_2. \tag{39}$$

The set of compatible CoPs is the singleton $\mathcal{C}_{\mathcal{M}_0}(y_0) = \{c[\nu_N^\star]\}$.

Theorem 4.3 (Exact coarse lower bound)**.** *Let $c = c[\nu_N^\star]$. Given the coarse observation $y_0 = (f_N, f_N c)$, for any $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\boxed{\underline{h}_{\mathcal{M}_0}(q; y_0) = \mu f_N \|a\|_2} \tag{40}$$

Proof. By Lemma 3.10(iii), $h_{\mathcal{W}(\nu_N)}(q) \geq \mu f_N \|a\|_2$ for every $\nu_N \in \Pi_{\mathcal{M}_0}(y_0)$. Since P is convex, $c \in P$, so $f_N \delta_c \in \Pi_{\mathcal{M}_0}(y_0)$, and it attains this value: the torque of a point load about its own location vanishes, so $\mathcal{W}(f_N \delta_c) = f_N B(0, \mu) \times \{0\}$. □

Remark 4.4. *As an immediate consequence, setting $a = 0$ shows that the coarse measurement does not guarantee any torsional capacity, i.e.* $\underline{h}_{\mathcal{M}_0}((0,b);y_0) = 0$.

Theorem 4.5 (Coarse upper bound)**.** *The supremum $\overline{h}_{\mathcal{M}_0}(q;y_0)$ is attained by a measure $\nu_N^+ \in \Pi_{\mathcal{M}_0}(y_0)$ supported on at most three points, i.e.,*

$$\nu_N^+ = \sum_{j=1}^{s} \lambda_j \delta_{x_j}, \qquad s \le 3, \tag{41}$$

where $x_j \in P$, $\lambda_j \ge 0$, and

$$\sum_{j=1}^{s} \lambda_j = f_N, \qquad \sum_{j=1}^{s} \lambda_j x_j = f_N c. \tag{42}$$

Proof. This is Proposition 4.4 of Yann's draft with the Coulomb set $C = B(0,\mu)$ (see Remark 4.7). There, as in [3], it follows from the fact that the extreme points of a set of measures defined by three linear moment constraints are supported on at most three points [2]. Check with Yann. □

Corollary 4.6 (Polygonal contact patch)**.** *Under the assumptions of Theorem 4.5,suppose additionally that P is a convex polygon with vertices $v_1,\dots,v_K$. Then the coarse upper bound is given exactly by the linear program*

$$\boxed{\begin{aligned} \overline{h}_{\mathcal{M}_0}(q;y_0) = \mu \max_{\lambda_1,\dots,\lambda_K} \quad & \sum_{k=1}^{K} \lambda_k \|a + bJ(v_k - c)\|_2 \\ s.t. \quad & \lambda_k \ge 0, \qquad k = 1,\dots,K, \\ & \sum_{k=1}^{K} \lambda_k = f_N, \\ & \sum_{k=1}^{K} \lambda_k v_k = f_N c. \end{aligned}} \tag{43}$$

Moreover, an optimal solution exists with at most three nonzero weights λ_k.

Proof. By Theorem 4.5, an optimal measure is $\sum_{j=1}^{s} \lambda_j \delta_{x_j}$ with $s \le 3$. Write each x_j as a convex combination $\sum_k \alpha_{jk} v_k$ of the vertices. Moving the mass $\lambda_j \alpha_{jk}$ from x_j to v_k preserves the zeroth and first moments and, by convexity of $g(\cdot;q,c)$ (Lemma 3.10(i)), does not decrease the objective. This is the reduction to the extreme points of P in Yann's Proposition 4.4. Hence an optimal measure supported on the vertices exists, which gives the LP. The LP has three equality constraints, so it has a basic optimal solution with at most three nonzero weights. □

Remark 4.7 (Connection with Yann's formulation To check with Yann)**.** *Theorem 4.5 and Corollary 4.6 are the Coulomb case $C = B(0,\mu)$ of Proposition 4.4 in Yann's draft, where the moment is taken about a fixed origin. With his notation $x^\perp = -Jx$ and a twist (ω_N, v_T), his integrand is $s_C(\omega_N x^\perp - v_T) = \mu\|v_T + \omega_N Jx\|_2$, which equals $\mu\, g(x;q,c)$ for*

$$b = \omega_N, \qquad a = v_T + \omega_N Jc,$$

Yann's $D(c^\perp)$ collects the moment and tangential force generated by all distributions with CoP c, so that $f_N\, s_{D(c^\perp)}(-[\omega_N, v_T]) = \overline{h}_{\mathcal{M}_0}(q;y_0)$ A possible nice interpretation is that $h_{\mathcal{W}(\nu_N)}(q)$ is the maximal power dissipated by the twist (MDP) and $\overline{h}_{\mathcal{M}_0}$ is its worst case over the distributions compatible with y_0.

The two bounds differ in general, which shows that the coarse measurement does not determine the admissible wrench set.

Example 4.8 (Segment patch). *Let* $P = [0, L] \times \{0\}$, $c = \left(\frac{L}{2}, 0\right)$, $\nu_N^{(1)} = f_N \delta_c$ *and* $\nu_N^{(2)} = \frac{f_N}{2}\delta_{(0,0)} + \frac{f_N}{2}\delta_{(L,0)}$. *Both have* $\mathcal{M}_0(\nu_N^{(1)}) = \mathcal{M}_0(\nu_N^{(2)}) = \left(f_N, f_N\left(\frac{L}{2}, 0\right)\right)$, *but in the pure torsion direction* $(a, b) = (0, 1)$,

$$h_{\mathcal{W}(\nu_N^{(1)})}(0,1) = 0 \quad \neq \quad h_{\mathcal{W}(\nu_N^{(2)})}(0,1) = \mu f_N \frac{L}{2} = \overline{h}_{\mathcal{M}_0}((0,1); y_0). \tag{44}$$

The results of this section are recovered in Section 6 as the special case of a single taxel $P_1 = P$.

5 Taxel measurements: preliminaries (common setting)

We now assume that a tactile array measures the normal force carried by each taxel, with or without the coarse measurement. This section fixes the setting and collects three lemmas used in Sections 6 and 7.

Assumption 5.1 (Taxel partition). *The patch* P *is partitioned into* r *nonempty, pairwise disjoint, convex Borel sets* $P_1, \dots, P_r$, *called taxels, with* $P = \bigcup_{i=1}^r P_i$. *When stated, the closure of each taxel is a convex polygon* $\overline{P_i} = \operatorname{conv}\{v_{i1}, \dots, v_{iK_i}\}$ *with vertices* $v_{ik} \in \mathbb{R}^2$.

Definition 5.2 (Taxel force measurement). *Given the true unknown normal force measure* $\nu_N^\star$ *and the partition* $\{P_i\}_{i=1}^r$, *the taxel force measurement is defined as*

$$y_F = \mathcal{M}_F(\nu_N^\star) \triangleq (\nu_N^\star(P_1), \dots, \nu_N^\star(P_r)). \tag{45}$$

Equivalently, its measurement functions are $\phi_i(x) = \mathbf{1}_{P_i}(x)$, $i = 1, \dots, r$. *The measured taxel forces are denoted* $F_i \triangleq \nu_N^\star(P_i)$, $i = 1, \dots, r$. *Because in practice we can not tell to which of two neighboring taxels an edge load belongs, we call a measure compatible with* y_F *if its force can be split among the taxel closures with the measured amounts:*

$$\Pi_{\mathcal{M}_F}(y_F) \triangleq \left\{\nu_N = \sum_{i=1}^r \nu_i : \ \nu_i \in \mathcal{M}_+(P) \text{ supported on } \overline{P_i},\ \nu_i(\overline{P_i}) = F_i,\ i = 1, \dots, r\right\}. \tag{46}$$

The set (46) slightly enlarges the preimage $\mathcal{M}_F^{-1}(y_F) = \{\nu_N : \nu_N(P_i) = F_i\ \forall i\}$ (take ν_i the restriction of ν_N to P_i); in particular it contains $\nu_N^\star$. The two sets differ only by loads on the taxel boundaries, which carry no force for a pressure density. In return, all the finite-dimensional formulas below hold with equality and are attained, without any argument on the taxel boundaries. To check with Yann.

Definition 5.3 (Combined coarse and taxel force measurement). *If both coarse and taxel force information are available, we define the combined coarse-and-taxel measurement*

$$y_T = \mathcal{M}_T(\nu_N^\star) \triangleq \left(\mathcal{M}_0(\nu_N^\star), \mathcal{M}_F(\nu_N^\star)\right) \tag{47}$$

$$= ((f_N, f_N c[\nu_N^\star]), F_1, \dots, F_r), \tag{48}$$

with compatible set $\Pi_{\mathcal{M}_T}(y_T) \triangleq \Pi_{\mathcal{M}_0}(y_0) \cap \Pi_{\mathcal{M}_F}(y_F)$.

Remark 5.4 (What is known). *Since the taxels form a partition of* P, y_F *determines the resultant normal force* $f_N = m_0(\nu_N^\star) = \sum_{i=1}^r F_i$, *and we assume* $f_N > 0$. *In both cases, the patch, the partition,* μ *and the taxel forces* F_i *are known, and the distribution of* $\nu_N^\star$ *inside each taxel is unknown. The CoP* $c^\star = c[\nu_N^\star]$ *is known under* $\mathcal{M}_T$, *but not determined by* y_F *under* $\mathcal{M}_F$. *With a single taxel* ($r = 1$, $P_1 = P$), $y_F = f_N$ *and* $\Pi_{\mathcal{M}_T}(y_T) = \Pi_{\mathcal{M}_0}(y_0)$: *the coarse case is the single-taxel case of* $\mathcal{M}_T$.

Lemma 5.5 (Local CoPs of the taxels). *Under Assumption 5.1, let $\nu_N = \sum_i \nu_i \in \Pi_{\mathcal{M}_F}(y_F)$ be split as in (46), with $f_N = \sum_i F_i > 0$. Define*

$$c_i \triangleq \begin{cases} \frac{1}{F_i} \int x \, d\nu_i(x) & \text{if } F_i > 0 \\ \text{any point in } \overline{P_i} & \text{if } F_i = 0, \end{cases} \tag{49}$$

i.e. c_i is the local CoP of the normal force carried by taxel i. Then (i) $c_i \in \overline{P_i}$; (ii) $c[\nu_N] = \frac{1}{f_N} \sum_{i=1}^r F_i c_i$, i.e. the CoP is the F_i-weighted average of the local CoPs; (iii) for every q and every $c \in \mathbb{R}^2$,

$$\int_P g(x; q, c) \, d\nu_N(x) \geq \sum_{i=1}^r F_i \, g(c_i; q, c). \tag{50}$$

Conversely, for any $c_i \in \overline{P_i}$, the measure $\nu_N = \sum_{i=1}^r F_i \delta_{c_i}$ belongs to $\Pi_{\mathcal{M}_F}(y_F)$, has CoP $\frac{1}{f_N} \sum_i F_i c_i$, and equality holds in (50).

Proof. (i) This is the same fact as $c \in P$ in the planar contact model (Lemma 4.1 of Yann's draft), applied to ν_i: a weighted average of points of the closed convex set $\overline{P_i}$ stays in it. For a rectangular taxel this is immediate coordinate-wise: each coordinate of c_i is an average of values in a closed interval, hence lies in it. (ii) $\int_P x \, d\nu_N = \sum_i \int x \, d\nu_i = \sum_i F_i c_i$. (iii) Lemma 3.10(iii) applied to each ν_i with $F_i > 0$, summed over i. The converse follows by taking $\nu_i = F_i \delta_{c_i}$. □

Proposition 5.6 (Compatible CoP set from taxel forces). *Under Assumption 5.1, the set of CoPs compatible with a taxel force observation $y_F = (F_1, \ldots, F_r)$ is the weighted Minkowski sum*

$$\mathcal{C}_{\mathcal{M}_F}(y_F) = \left\{ \frac{1}{f_N} \sum_{i=1}^r F_i c_i : c_i \in \overline{P_i}, \quad i = 1, \ldots, r \right\} = \sum_{i=1}^r \frac{F_i}{f_N} \overline{P_i}, \tag{51}$$

which is compact and convex, and $\operatorname{diam} \mathcal{C}_{\mathcal{M}_F}(y_F) \leq \frac{1}{f_N} \sum_{i=1}^r F_i \operatorname{diam}(P_i) \leq \max_i \operatorname{diam}(P_i)$.

Proof. The inclusion $\subseteq$ follows from Lemma 5.5(i)–(ii), and $\supseteq$ from its converse part. For the diameter, let $c = \frac{1}{f_N} \sum_i F_i c_i$ and $c' = \frac{1}{f_N} \sum_i F_i c_i'$ with $c_i, c_i' \in \overline{P_i}$. By the triangle inequality,

$$\|c - c'\|_2 \leq \frac{1}{f_N} \sum_{i=1}^r F_i \|c_i - c_i'\|_2 \leq \frac{1}{f_N} \sum_{i=1}^r F_i \operatorname{diam}(P_i) \leq \max_i \operatorname{diam}(P_i), \tag{52}$$

where the last inequality uses $F_i \geq 0$ and $\sum_i F_i = f_N$. □

Remark 5.7 (Intuition Discuss with Yann). *In order to reduce infinite dimensional optimization over measures to finite dimensional optimization problems we need some lemmas/machinery (Figure 2).*

- *Lower bound = "concentrate". Collapsing the force of each taxel onto its local CoP keeps F_i and the CoP, and cannot increase the objective, since a bowl evaluated at an average point is below the average of the bowl (Jensen, Lemma 5.5(iii)). One point per taxel suffices. Needed for all lower bounds, $\mathcal{M}_0$ (Theorem 4.3, a single point at the CoP), $\mathcal{M}_T$ (Theorem 6.1) and $\mathcal{M}_F$ (Theorem 7.2).*
- *Upper bound = "spread to corners". Splitting a point load among the corners of its taxel, with weights whose average position is the load's position, keeps F_i and the CoP, and cannot decrease the objective, since a bowl lies below its chords. Every compatible measure is thus dominated by an LP over corner weights (Lemma 5.8, proved by LP duality). This is Yann's reduction to the extreme points of the patch (Corollary 4.6), applied taxel by taxel; Example 4.8 is the simplest instance. Needed for all upper bounds, $\mathcal{M}_0$ (Corollary 4.6, corners of P), $\mathcal{M}_T$ (Theorem 6.2) and $\mathcal{M}_F$ (Theorem 7.3).*

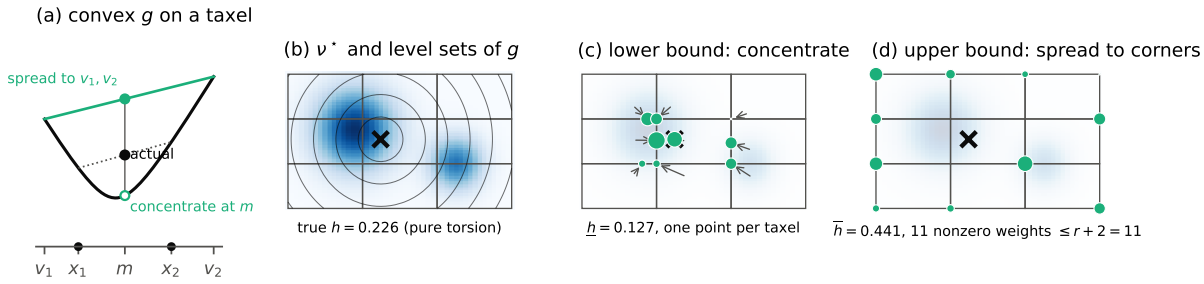


Figure 2: Concentrate vs. spread (Remark 5.7), 3×3 taxels, pure torsion $q = (0, 0, 1)$, so $g(x; q, c) = \|x - c\|_2$. (a) For masses at x_1, x_2 with CoP m: concentrating at m lowers the objective (Jensen), spreading to the vertices v_1, v_2 raises it (chord). (b) True measure and level sets of g. (c) Minimizer of Theorem 6.1: one Dirac per taxel, pulled from the local CoPs of the taxels toward the CoP (arrows). (d) Maximizer of Theorem 6.2: weights on the taxel corners, pushed away from the CoP, with at most $r + 2$ nonzero weights. Marker areas are proportional to force.

- *Boundaries. A corner of a half open taxel may belong to its neighbor. In our model (Definition 5.2), a load on a shared edge or corner may be counted by either taxel, so corner-supported measures are admissible and all the finite-dimensional problems below are exact. See Remark 5.9.*

Lemma 5.8 (Vertex reduction)**.** *Under Assumption 5.1 with polygonal taxel closures, let* $\nu_N = \sum_i \nu_i \in \Pi_{\mathcal{M}_F}(y_F)$*, with* $f_N = \sum_i F_i > 0$ *and* $\int_P x \, d\nu_N(x) = f_N c$*. Then, for every* q*,*

$$\int_P g(x; q, c)\, d\nu_N(x) \le V(q, c) \triangleq \max_{\lambda_{ik} \ge 0} \left\{ \sum_{i,k} \lambda_{ik}\, g(v_{ik}; q, c) : \sum_{k=1}^{K_i} \lambda_{ik} = F_i \ \forall i, \quad \sum_{i,k} \lambda_{ik} v_{ik} = f_N c \right\}. \tag{53}$$

Proof. We follow the idea of Yann's draft (Section 5.2) which uses the dual of the same LP for the whole patch (i.e., case $r = 1$). *The LP is feasible and bounded:* By Lemma 5.5, the local CoPs $c_i \in \overline{P_i}$ satisfy $\sum_i F_i c_i = f_N c$. Writing each c_i as a convex combination $\sum_k \alpha_{ik} v_{ik}$ of the corners of its taxel, $\lambda_{ik} = F_i \alpha_{ik}$ is feasible. The feasible set is compact, so the maximum exists.

Dual LP: With multipliers y_i for the taxel constraints and $z \in \mathbb{R}^2$ for the CoP constraint, LP duality gives

$$V(q, c) = \min_{y \in \mathbb{R}^r,\, z \in \mathbb{R}^2} \left\{ \sum_{i=1}^{r} F_i y_i + f_N\, z^\top c : \ y_i + z^\top v_{ik} \ge g(v_{ik}; q, c) \ \forall i, k \right\}.$$

Affine upper bound on each taxel: Fix a dual-feasible (y, z), and let $\ell_i(x) = y_i + z^\top x$. By the constraints, $\ell_i \ge g(\cdot; q, c)$ at the corners of $\overline{P_i}$. A convex function that lies below an affine one at the corners of a polygon lies below it on the whole polygon: for $x = \sum_k \alpha_k v_{ik}$, $g(x; q, c) \le \sum_k \alpha_k g(v_{ik}; q, c) \le \sum_k \alpha_k \ell_i(v_{ik}) = \ell_i(x)$ (Lemma 3.10(i)).

Integrate: Since ν_i carries F_i on $\overline{P_i}$ and ν_N has first moment $f_N c$,

$$\int_P g(x; q, c)\, d\nu_N(x) = \sum_{i=1}^{r} \int g(x; q, c)\, d\nu_i(x) \le \sum_{i=1}^{r} \int \left(y_i + z^\top x \right) d\nu_i(x) = \sum_{i=1}^{r} F_i y_i + f_N\, z^\top c.$$

Then we take the min over dual feasible (y, z) and get (53). □

Remark 5.9 (Strict half-open taxels)**.** *One could instead require $\nu_N(P_i) = F_i$ exactly, with the half-open taxels, i.e., use the preimage $\mathcal{M}_F^{-1}(y_F) \subseteq \Pi_{\mathcal{M}_F}(y_F)$. Our bounds are conservative for this stricter model (Lemma 3.12). If we adopted the stricte convention we could still construct some admissible distribution by pushing the corner points toward the interior in a way that does not affect the CoP (important for $\mathcal{M}_T$ which constrains it)?*

6 $\mathcal{M}_T$ case (taxel forces with CoP)

Known: the taxel forces F_i and the CoP $c^\star = c[\nu_N^\star]$. **Unknown:** the distribution of $\nu_N^\star$ inside each taxel. Since y_T determines the CoP, the set of compatible CoPs is reduced to a singleton $\mathcal{C}_{\mathcal{M}_T}(y_T) = \{c^\star\}$. Hence the outer optimization of Lemma 3.11 disappears, and the $\mathcal{M}_T$-conditioned bounds reduce to the GMPs

$$\underline{h}_{\mathcal{M}_T}(q; y_T) = \inf_{\substack{\nu_N \in \Pi_{\mathcal{M}_F}(y_F) \\ \int_P (x-c^\star)\, d\nu_N = 0}} \mu \int_P \|a + bJ(x - c^\star)\|_2 \, d\nu_N(x), \tag{54}$$

$$\overline{h}_{\mathcal{M}_T}(q; y_T) = \sup_{\substack{\nu_N \in \Pi_{\mathcal{M}_F}(y_F) \\ \int_P (x-c^\star)\, d\nu_N = 0}} \mu \int_P \|a + bJ(x - c^\star)\|_2 \, d\nu_N(x). \tag{55}$$

The taxel force constraints (46) act on each taxel separately, but the CoP consistency constraint $\int_P (x - c^\star)\, d\nu_N = 0$ couples the measure ν_N across the taxels. In what follows, c denotes the measured CoP $c^\star$.

Theorem 6.1 (Exact lower bound for combined measurements)**.** *Under Assumption 5.1, given the combined observation*

$$y_T = \mathcal{M}_T(\nu_N^\star) = (y_0, y_F), \tag{56}$$

$$y_0 = (f_N, f_N c), \qquad y_F = (F_1, \ldots, F_r), \tag{57}$$

where $f_N = \sum_{i=1}^r F_i > 0$, the measurement-conditioned lower bound satisfies

$$\boxed{\begin{aligned} \underline{h}_{\mathcal{M}_T}(q; y_T) = \mu \min_{c_1,\ldots,c_r} \quad & \sum_{i=1}^r F_i \, \|a + bJ(c_i - c)\|_2 \\ \text{subject to} \quad & c_i \in \overline{P_i}, \qquad i = 1, \ldots, r, \\ & \sum_{i=1}^r F_i c_i = f_N c. \end{aligned}} \tag{58}$$

The resulting finite-dimensional optimization problem is convex, and an SOCP when the taxel closures are polygons.

Proof. This is the "concentrate" step of Remark 5.7. ($\geq$) For $\nu_N \in \Pi_{\mathcal{M}_T}(y_T)$, its local CoPs are feasible and $h_{\mathcal{W}(\nu_N)}(q) \geq \mu \sum_i F_i \, g(c_i; q, c)$, by Lemma 5.5. ($\leq$) For feasible (c_i), the measure $\sum_i F_i \delta_{c_i} \in \Pi_{\mathcal{M}_T}(y_T)$ attains the objective (converse part of Lemma 5.5). The minimum exists (compact feasible set, nonempty since $c \in \mathcal{C}_{\mathcal{M}_F}(y_F)$, continuous objective), and the problem is convex by Lemma 3.10(i), since the constraints are convex and affine. For polygonal closures, epigraph variables $t_i \geq \|a + bJ(c_i - c)\|_2$ give an SOCP. □

Theorem 6.2 (Exact upper bound for combined measurements)**.** *Under the assumptions of Theorem 6.1, suppose additionally that each taxel closure is a polygon,*

$$\overline{P_i} = \operatorname{conv}\{v_{i1}, \dots, v_{iK_i}\}, \tag{59}$$

where $v_{ik} \in \mathbb{R}^2$ are its vertices. Then, for every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,

$$\boxed{\begin{aligned} \overline{h}_{\mathcal{M}_T}(q; y_T) = \mu \max_{\lambda_{ik} \geq 0} \quad & \sum_{i=1}^{r} \sum_{k=1}^{K_i} \lambda_{ik} \left\| a + bJ(v_{ik} - c) \right\|_2 \\ \text{subject to} \quad & \sum_{k=1}^{K_i} \lambda_{ik} = F_i, \qquad i = 1, \dots, r, \\ & \sum_{i=1}^{r} \sum_{k=1}^{K_i} \lambda_{ik} v_{ik} = f_N c. \end{aligned}} \tag{60}$$

The resulting finite-dimensional optimization problem is a linear program, and an optimal solution exists with at most $r + 2$ nonzero weights. In particular, the supremum is attained by a corner-supported measure.

Proof. This is the "spread to corners" step of Remark 5.7. ($\leq$) For every $\nu_N \in \Pi_{\mathcal{M}_T}(y_T)$, $h_{\mathcal{W}(\nu_N)}(q) \leq \mu V(q, c)$ by Lemma 5.8, and $V(q, c)$ is the LP (60). ($\geq$) For an optimal λ, the corner-supported measure $\sum_{i,k} \lambda_{ik} \delta_{v_{ik}}$ belongs to $\Pi_{\mathcal{M}_T}(y_T)$ (its piece on $\overline{P_i}$ is $\sum_k \lambda_{ik} \delta_{v_{ik}}$) and attains the LP value. The LP is feasible and bounded (proof of Lemma 5.8), and a basic optimal solution has at most as many nonzero entries as equality constraints, i.e., $r + 2$. □

Remark 6.3 (Recovering the coarse case)**.** *For a single taxel $P_1 = P$, the constraint of (58) forces $c_1 = c$, and Theorem 6.1 reduces to Theorem 4.3. Theorem 6.2 reduces to Corollary 4.6, with $r + 2 = 3$ nonzero weights.*

The exact $\mathcal{M}_T$-conditioned bounds retain both the taxel-force constraints and the global first-moment constraint associated with the measured CoP. Simpler bounds can be obtained by treating the taxels independently. For a reference point $c \in \mathbb{R}^2$ and $i = 1, \dots, r$, define

$$g_i^{\min}(q; c) \triangleq \min_{x \in \overline{P_i}} \left\| a + bJ(x - c) \right\|_2, \tag{61}$$

$$g_i^{\max}(q; c) \triangleq \max_{x \in \overline{P_i}} \left\| a + bJ(x - c) \right\|_2. \tag{62}$$

For $b \neq 0$, $a + bJ(x - c) = bJ(x - p)$ with $p = c + Ja/b$, the instantaneous center of rotation. Hence $g_i^{\min}(q; c) = |b| \operatorname{dist}(p, P_i)$ and, for polygonal closures, $g_i^{\max}(q; c) = |b| \max_k \|v_{ik} - p\|_2$. For $b = 0$, both equal $\|a\|_2$.

Proposition 6.4 (Constructive cellwise support bounds)**.** *Let $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$. Then, for every $\nu_N \in \Pi_{\mathcal{M}_T}(y_T)$,*

$$\boxed{\mu \sum_{i=1}^{r} F_i g_i^{\min}(q; c^\star) \leq h_{\mathcal{W}(\nu_N)}(q) \leq \mu \sum_{i=1}^{r} F_i g_i^{\max}(q; c^\star).} \tag{63}$$

Proof. For every $x \in \overline{P_i}$,

$$g_i^{\min}(q; c^\star) \leq \left\| a + bJ(x - c^\star) \right\|_2 \leq g_i^{\max}(q; c^\star). \tag{64}$$

Write $\nu_N = \sum_i \nu_i$ as in (46). Since ν_i is nonnegative with mass F_i on $\overline{P_i}$, integration against ν_i gives

$$F_i g_i^{\min}(q; c^\star) \le \int \|a + bJ(x - c^\star)\|_2 \, d\nu_i(x) \le F_i g_i^{\max}(q; c^\star). \tag{65}$$

Summing over i yields

$$\sum_{i=1}^{r} F_i g_i^{\min}(q; c^\star) \le \int_P \|a + bJ(x - c^\star)\|_2 \, d\nu_N(x) \le \sum_{i=1}^{r} F_i g_i^{\max}(q; c^\star). \tag{66}$$

The result follows from Proposition 3.2. □

Consequently,

$$\boxed{\mu \sum_{i=1}^{r} F_i g_i^{\min}(q; c^\star) \le \underline{h}_{\mathcal{M}_T}(q; y_T) \le \overline{h}_{\mathcal{M}_T}(q; y_T) \le \mu \sum_{i=1}^{r} F_i g_i^{\max}(q; c^\star).} \tag{67}$$

These cellwise bounds are directly computable from the taxel forces, the measured CoP, and the taxel geometry, but need not be sharp. In particular, independently attaining the minimum or maximum within each taxel may violate the global first-moment constraint contained in $\mathcal{M}_T$.

7 $\mathcal{M}_F$ case (taxel forces without CoP)

Known: the taxel forces F_i. **Unknown:** the CoP $c^\star$ and the distribution of $\nu_N^\star$ inside each taxel. Contrary to y_0 and y_T, the taxel force measurement y_F does not determine the CoP. It only confines it to the convex set $\mathcal{C}_{\mathcal{M}_F}(y_F)$ of Proposition 5.6, which shrinks as the taxel partition is refined. The bounds therefore keep the bi-level structure of Lemma 3.11, and the inner problems are exactly the $\mathcal{M}_T$ problems of Section 6.

Proposition 7.1 ($\mathcal{M}_F$ as $\mathcal{M}_T$ with unknown CoP)**.** *For $c \in \mathcal{C}_{\mathcal{M}_F}(y_F)$, let $y_T(c) \triangleq ((f_N, f_N c), y_F)$. Then $\Pi_{\mathcal{M}_F}(y_F) = \bigcup_{c \in \mathcal{C}_{\mathcal{M}_F}(y_F)} \Pi_{\mathcal{M}_T}(y_T(c))$, and for every q,*

$$\boxed{\underline{h}_{\mathcal{M}_F}(q; y_F) = \inf_{c \in \mathcal{C}_{\mathcal{M}_F}(y_F)} \underline{h}_{\mathcal{M}_T}(q; y_T(c)), \qquad \overline{h}_{\mathcal{M}_F}(q; y_F) = \sup_{c \in \mathcal{C}_{\mathcal{M}_F}(y_F)} \overline{h}_{\mathcal{M}_T}(q; y_T(c)).} \tag{68}$$

Proof. With the notation of Lemma 3.11, $\Pi_{\mathcal{M}_F}(y_F; c) = \Pi_{\mathcal{M}_T}(y_T(c))$. □

Proposition 7.1 is a nested problem: an outer optimization over the unknown CoP $c \in \mathcal{C}_{\mathcal{M}_F}(y_F)$, and inner $\mathcal{M}_T$ problems at each c (Theorems 6.1 and 6.2). For both bounds, the outer variable can be eliminated exactly: the CoP constraint of the inner problem determines c from the inner decision variables, as the CoP $\bar{c}$ of the candidate measure. This gives the single-level problems of Theorems 7.2 and 7.3. They differ in convexity: the lower bound is convex, the upper bound is generally not. For the upper bound, fixing c gives back the LP (60), so a possible numerical strategy is an outer search of LPs over $c \in \mathcal{C}_{\mathcal{M}_F}(y_F)$.

Theorem 7.2 (Exact lower bound for taxel force measurements)**.** *Under Assumption 5.1, let $y_F = (F_1, \dots, F_r) = \mathcal{M}_F(\nu_N^\star)$, where $F_i = \nu_N^\star(P_i) \ge 0$, and assume that $f_N = \sum_{i=1}^r F_i > 0$. Then, for every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\boxed{\underline{h}_{\mathcal{M}_F}(q; y_F) = \mu \min_{\substack{c_i \in \overline{P_i} \\ i=1,\dots,r}} \sum_{i=1}^{r} F_i \, \|a + bJ(c_i - \bar{c})\|_2} \tag{69}$$

where

$$\bar{c} \triangleq \frac{1}{f_N} \sum_{i=1}^{r} F_i c_i \tag{70}$$

is the CoP of the candidate measure $\sum_i F_i \delta_{c_i}$. This finite-dimensional optimization problem is convex, and an SOCP when the taxel closures are polygons.

Proof. By (68) and Theorem 6.1, minimizing (58) jointly over $c \in \mathcal{C}_{\mathcal{M}_F}(y_F)$ and the c_i eliminates the constraint $\sum_i F_i c_i = f_N c$, with $c = \bar{c}$. Equivalently, directly: for $\nu_N \in \Pi_{\mathcal{M}_F}(y_F)$, Lemma 5.5 with $c = c[\nu_N] = \bar{c}$ gives the inequality $\geq$, and the measures $\sum_i F_i \delta_{c_i}$ give the converse.

Since $\bar{c}$ is affine in $(c_1, \ldots, c_r)$, the objective is convex. The minimum exists (compact feasible set, continuous objective), and for polygonal closures epigraph variables give an SOCP. □

Theorem 7.3 (Exact upper bound for polygonal taxels)**.** *Under the assumptions of Theorem 7.2, suppose additionally that each taxel closure is a polygon $\overline{P_i} = \operatorname{conv}\{v_{i1}, \ldots, v_{iK_i}\}$ where $v_{ik} \in \mathbb{R}^2$ are its vertices. Let $\lambda_{ik} \geq 0$ such that $\sum_{k=1}^{K_i} \lambda_{ik} = F_i$, define*

$$\bar{c}(\lambda) \triangleq \frac{1}{f_N} \sum_{i=1}^{r} \sum_{k=1}^{K_i} \lambda_{ik} v_{ik}, \tag{71}$$

the CoP of the candidate measure $\sum_{i,k} \lambda_{ik} \delta_{v_{ik}}$ (again a function of the decision variables, not a measurement). Then, for every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,

$$\begin{aligned} \overline{h}_{\mathcal{M}_F}(q; y_F) = \mu \max_{\lambda_{ik} \geq 0} \quad & \sum_{i=1}^{r} \sum_{k=1}^{K_i} \lambda_{ik} \left\| a + bJ\big(v_{ik} - \bar{c}(\lambda)\big) \right\|_2 \\ \text{subject to} \quad & \sum_{k=1}^{K_i} \lambda_{ik} = F_i, \qquad i = 1, \ldots, r. \end{aligned} \tag{72}$$

This single-level problem is generally nonconvex: the weights λ_{ik} multiply norms that depend on λ through $\bar{c}(\lambda)$. Fixing $c = \bar{c}(\lambda)$ turns it into the LP (60), so $\overline{h}_{\mathcal{M}_F}(q; y_F)$ is the maximum over $c \in \mathcal{C}_{\mathcal{M}_F}(y_F)$ of this LP. A finite sweep over c therefore gives a value $\leq \overline{h}_{\mathcal{M}_F}(q; y_F)$, which is not certified; Proposition 7.4 gives a certified value $\geq \overline{h}_{\mathcal{M}_F}(q; y_F)$.

Proof. ($\leq$) For $\nu_N \in \Pi_{\mathcal{M}_F}(y_F)$ with CoP $c = c[\nu_N]$, Lemma 5.8 gives $h_{\mathcal{W}(\nu_N)}(q) \leq \mu V(q, c)$, and every λ feasible for $V(q, c)$ is feasible for (72), with $\bar{c}(\lambda) = c$. ($\geq$) For feasible λ, the corner-supported measure $\sum_{i,k} \lambda_{ik} \delta_{v_{ik}}$ belongs to $\Pi_{\mathcal{M}_F}(y_F)$, has CoP $\bar{c}(\lambda)$, and attains the objective of λ. The maximum exists, since the feasible set is compact and the objective continuous. Finally, grouping the feasible λ by the value of $\bar{c}(\lambda) \in \sum_i \frac{F_i}{f_N} \overline{P_i} = \mathcal{C}_{\mathcal{M}_F}(y_F)$ (Proposition 5.6) gives the equivalent form, i.e., Proposition 7.1 with the LP (60) as inner problem. □

Fixed-reference bounds. In practice, an estimate $\hat{c}$ of the CoP is often available. It can be used in two different ways. *Fixing the CoP*, i.e., imposing $\int_P (x - \hat{c})\, d\nu_N = 0$, gives a single $\mathcal{M}_T$ problem; this is valid only if $\hat{c}$ is the true CoP, and otherwise it is just one slice of Proposition 7.1. *Fixing the reference* instead means using $\hat{c}$ only as the point about which the torque is computed, without enforcing the CoP constraint. In that case, the taxel constraints are independent and the optimization decouples into the cellwise quantities of Section 6:

$$\underline{h}_F^{\mathrm{ref}}(q; y_F, \hat{c}) = \mu \sum_{i=1}^{r} F_i\, g_i^{\min}(q; \hat{c}), \qquad \overline{h}_F^{\mathrm{ref}}(q; y_F, \hat{c}) = \mu \sum_{i=1}^{r} F_i\, g_i^{\max}(q; \hat{c}). \tag{73}$$

These expressions are not bounds for the $\mathcal{M}_F$-conditioned problem, because the true CoP differs from $\hat{c}$. They become bounds after a correction for the uncertainty of the CoP.

Proposition 7.4 (Certified fixed-reference bounds)**.** *For every $\hat{c} \in \mathbb{R}^2$, let $\delta(\hat{c}) \triangleq \sup_{c \in \mathcal{C}_{\mathcal{M}_F}(y_F)} \|c - \hat{c}\|_2$. Then, for every $q = (a, b)$,*

$$\underline{h}_F^{\mathrm{ref}}(q; y_F, \hat{c}) - \mu|b| f_N \delta(\hat{c}) \le \underline{h}_{\mathcal{M}_F}(q; y_F) \le \overline{h}_{\mathcal{M}_F}(q; y_F) \le \overline{h}_F^{\mathrm{ref}}(q; y_F, \hat{c}) + \mu|b| f_N \delta(\hat{c}). \tag{74}$$

For $\hat{c} \in \mathcal{C}_{\mathcal{M}_F}(y_F)$, $\delta(\hat{c}) \le \operatorname{diam} \mathcal{C}_{\mathcal{M}_F}(y_F)$.

Proof. Let $\nu_N \in \Pi_{\mathcal{M}_F}(y_F)$ with $c = c[\nu_N] \in \mathcal{C}_{\mathcal{M}_F}(y_F)$. By Lemma 3.10(ii), $|g(x; q, c) - g(x; q, \hat{c})| \le |b|\delta(\hat{c})$, so $\left| h_{\mathcal{W}(\nu_N)}(q) - \mu \int_P g(x; q, \hat{c})\, d\nu_N \right| \le \mu|b| f_N \delta(\hat{c})$. The cellwise argument of Proposition 6.4, applied with reference $\hat{c}$, bounds $\mu \int_P g(x; q, \hat{c})\, d\nu_N$ between $\underline{h}_F^{\mathrm{ref}}$ and $\overline{h}_F^{\mathrm{ref}}$. □

Remark 7.5 (Using an estimate of the CoP)**.** *The correction $\delta(\hat{c})$ is a supremum over all of $\mathcal{C}_{\mathcal{M}_F}(y_F)$, so a more accurate $\hat{c}$ does not reduce it by itself; it is smallest when $\hat{c}$ is the centre of $\mathcal{C}_{\mathcal{M}_F}(y_F)$. If additional information guarantees that the true CoP lies in a subset $\mathcal{C} \subseteq \mathcal{C}_{\mathcal{M}_F}(y_F)$ (e.g. an estimate with a known error radius), the same proof applies to the measures with CoP in $\mathcal{C}$, with $\delta(\hat{c})$ taken over $\mathcal{C}$. For the upper bound, Proposition 7.4 avoids the nonconvex problem of Theorem 7.3. The exact upper bound is then bracketed: the best value found by a finite sweep over $c \in \mathcal{C}_{\mathcal{M}_F}(y_F)$ of the LP (60) is $\le \overline{h}_{\mathcal{M}_F}(q; y_F) \le \overline{h}_F^{\mathrm{ref}}(q; y_F, \hat{c}) + \mu|b| f_N \delta(\hat{c})$. For the lower bound, Theorem 7.2 is already a single convex program and at least as tight, so an estimate of the CoP is not needed there.*

8 Comparison of measurements and tightening under refinement

All results of this section follow from Lemma 3.12 (inclusion of compatible families) and from the Lipschitz property of Lemma 3.10(ii).

Proposition 8.1 (Comparison of coarse, taxel and combined measurements)**.** *The compatible measure family associated with the combined measurement is*

$$\Pi_{\mathcal{M}_T}(y_T) = \Pi_{\mathcal{M}_0}(y_0) \cap \Pi_{\mathcal{M}_F}(y_F). \tag{75}$$

Consequently, for every wrench direction $q \in \mathbb{R}^2 \times \mathbb{R}$,

$$\boxed{\mu f_N \|a\|_2 = \underline{h}_{\mathcal{M}_0}(q; y_0) \le \underline{h}_{\mathcal{M}_F}(q; y_F) \le \underline{h}_{\mathcal{M}_T}(q; y_T) \le h_{\mathcal{W}(\nu_N^\star)}(q)} \tag{76}$$

and

$$\boxed{h_{\mathcal{W}(\nu_N^\star)}(q) \le \overline{h}_{\mathcal{M}_T}(q; y_T) \le \min\left\{\overline{h}_{\mathcal{M}_0}(q; y_0), \overline{h}_{\mathcal{M}_F}(q; y_F)\right\}.} \tag{77}$$

Thus, $\mathcal{M}_T$ provides bounds that are at least as tight as those obtained from either $\mathcal{M}_0$ or $\mathcal{M}_F$ alone. The lower bounds of $\mathcal{M}_0$ and $\mathcal{M}_F$ are ordered, whereas their upper bounds are not in general ordered.

Proof. The identity follows from $y_T = (y_0, y_F)$, and the inequalities involving $\mathcal{M}_T$ from Lemma 3.12. Every $\nu_N \in \Pi_{\mathcal{M}_F}(y_F)$ has total mass f_N, so (34) gives $\underline{h}_{\mathcal{M}_F}(q; y_F) \ge \mu f_N \|a\|_2 = \underline{h}_{\mathcal{M}_0}(q; y_0)$ (Theorem 4.3). For the upper bounds, with a single taxel $\Pi_{\mathcal{M}_F}(y_F) \supseteq \Pi_{\mathcal{M}_0}(y_0)$, so $\overline{h}_{\mathcal{M}_F} \ge \overline{h}_{\mathcal{M}_0}$. For fine partitions, $\overline{h}_{\mathcal{M}_F} \to h_{\mathcal{W}(\nu_N^\star)}$ by Proposition 8.3, which is generally smaller than $\overline{h}_{\mathcal{M}_0}$ (Corollary **??**). □

Proposition 8.2 (Tightening under taxel refinement)**.** *Let $\mathcal{M}'_F$ and $\mathcal{M}'_T$ denote the taxel-force and combined measurements associated with a partition $\{P'_j\}$ that refines the partition $\{P_i\}$ associated with $\mathcal{M}_F$ and $\mathcal{M}_T$ (each P_i is a union of cells P'_j), and let*

$$y_F = \mathcal{M}_F(\nu_N^\star), \quad y'_F = \mathcal{M}'_F(\nu_N^\star), \quad y_T = \mathcal{M}_T(\nu_N^\star), \quad y'_T = \mathcal{M}'_T(\nu_N^\star). \tag{78}$$

Then

$$\Pi_{\mathcal{M}'_F}(y'_F) \subseteq \Pi_{\mathcal{M}_F}(y_F), \qquad \Pi_{\mathcal{M}'_T}(y'_T) \subseteq \Pi_{\mathcal{M}_T}(y_T), \qquad \mathcal{C}_{\mathcal{M}'_F}(y'_F) \subseteq \mathcal{C}_{\mathcal{M}_F}(y_F). \tag{79}$$

Consequently, for every $q \in \mathbb{R}^2 \times \mathbb{R}$ and $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$, with y and y' the corresponding observations,

$$\boxed{\underline{h}_{\mathcal{M}}(q; y) \le \underline{h}_{\mathcal{M}'}(q; y') \le h_{\mathcal{W}(\nu_N^\star)}(q) \le \overline{h}_{\mathcal{M}'}(q; y') \le \overline{h}_{\mathcal{M}}(q; y).} \tag{80}$$

Proof. If $\nu_N = \sum_j \nu'_j \in \Pi_{\mathcal{M}'_F}(y'_F)$, grouping the pieces inside each taxel, $\nu_i = \sum_{j: P'_j \subseteq P_i} \nu'_j$, gives measures supported on $\overline{P_i}$ with mass $\sum_{j: P'_j \subseteq P_i} \nu_N^\star(P'_j) = F_i$, so $\nu_N \in \Pi_{\mathcal{M}_F}(y_F)$. Intersecting with $\Pi_{\mathcal{M}_0}(y_0)$ gives the inclusion for $\mathcal{M}_T$ (Proposition 8.1). Conclude with Lemma 3.12. □

The monotonicity above is associated with *partition refinement*, not with the number of taxels r alone. The convergence below, on the other hand, only requires the taxel diameters to vanish.

Proposition 8.3 (Resolution-dependent gap bounds)**.** *For every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\boxed{0 \le \overline{h}_{\mathcal{M}_T}(q; y_T) - \underline{h}_{\mathcal{M}_T}(q; y_T) \le \mu|b| \sum_{i=1}^{r} F_i \operatorname{diam}(P_i) \le \mu|b| f_N \max_i \operatorname{diam}(P_i),} \tag{81}$$

$$\boxed{0 \le \overline{h}_{\mathcal{M}_F}(q; y_F) - \underline{h}_{\mathcal{M}_F}(q; y_F) \le 2\mu|b| \sum_{i=1}^{r} F_i \operatorname{diam}(P_i) \le 2\mu|b| f_N \max_i \operatorname{diam}(P_i).} \tag{82}$$

Hence, along any sequence of taxel partitions with $\max_i \operatorname{diam}(P_i) \to 0$,

$$\underline{h}_{\mathcal{M}_F}(q; y_F),\ \underline{h}_{\mathcal{M}_T}(q; y_T),\ \overline{h}_{\mathcal{M}_T}(q; y_T),\ \overline{h}_{\mathcal{M}_F}(q; y_F) \to h_{\mathcal{W}(\nu_N^\star)}(q). \tag{83}$$

Proof. $\mathcal{M}_T$: by (67), the gap is at most $\mu \sum_i F_i (g_i^{\max} - g_i^{\min})(q; c^\star)$, and $g_i^{\max} - g_i^{\min} \le |b| \operatorname{diam}(P_i)$ by Lemma 3.10(ii). $\mathcal{M}_F$: let $\nu_N, \nu'_N \in \Pi_{\mathcal{M}_F}(y_F)$, with CoPs $c, c' \in \mathcal{C}_{\mathcal{M}_F}(y_F)$. For $x, x' \in \overline{P_i}$, Lemma 3.10(ii) gives $g(x; q, c) - g(x'; q, c') \le |b| \,(\operatorname{diam}(P_i) + \operatorname{diam} \mathcal{C}_{\mathcal{M}_F}(y_F))$. Integrating against the pieces of ν_N and ν'_N on each taxel,

$$h_{\mathcal{W}(\nu_N)}(q) - h_{\mathcal{W}(\nu'_N)}(q) \le \mu|b| \left(\sum_{i=1}^{r} F_i \operatorname{diam}(P_i) + f_N \operatorname{diam} \mathcal{C}_{\mathcal{M}_F}(y_F) \right) \le 2\mu|b| \sum_{i=1}^{r} F_i \operatorname{diam}(P_i), \tag{84}$$

by Proposition 5.6. Take the supremum over ν_N and the infimum over ν'_N. Convergence follows since $h_{\mathcal{W}(\nu_N^\star)}(q)$ lies between the bounds (Lemma 3.12). □

9 Certified load factor

A controller does not need the bounds direction by direction. It needs to know whether a given load w can be carried and, if not, by how much the contact should be strengthened (e.g., by squeezing harder). This is a question about the ray $\{sw : s \geq 0\}$: how far along it the guaranteed set extends. This section shows that the answer is one convex program for the taxel measurements, without constructing the guaranteed set.

Fixed reference. The load acting on a grasped object is known about a point o fixed to the object (or to the hand), not about the CoP, which is unknown under $\mathcal{M}_F$. We therefore take the torque about a fixed point $o \in \mathbb{R}^2$.

Definition 9.1 (Admissible tangential wrench set about a fixed point)**.** *For $\nu_N \in \mathcal{M}_+(P)$ and $o \in \mathbb{R}^2$, let $\mathcal{W}_o(\nu_N)$ be the set of (f_T, τ_o) with $f_T = \int_P \gamma_T \, d\nu_N$ and $\tau_o = \int_P (x-o) \times \gamma_T(x) \, d\nu_N(x)$, over all measurable γ_T with $\|\gamma_T\|_2 \leq \mu$ ν_N-a.e. Since $\tau_o = \tau_N + (c[\nu_N] - o) \times f_T$, $\mathcal{W}_o(\nu_N)$ is the image of $\mathcal{W}(\nu_N)$ by an invertible linear map; in particular it is convex, compact and contains 0, and $\mathcal{W}_{c[\nu_N]}(\nu_N) = \mathcal{W}(\nu_N)$. The proof of Proposition 3.2, with $c[\nu_N]$ replaced by o, gives its support function*

$$h_{\mathcal{W}_o(\nu_N)}(q) = \mu \int_P g(x; q, o) \, d\nu_N(x). \tag{85}$$

As in Definition 3.7 and Remark 3.9, we write

$$\underline{h}^o_{\mathcal{M}}(q; y) \triangleq \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{W}_o(\nu_N)}(q), \qquad \mathcal{W}^-_o(y) \triangleq \bigcap_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{W}_o(\nu_N). \tag{86}$$

The integrand $g(x; q, o)$ does not depend on the CoP of ν_N. The lower bounds are therefore GMPs for every measurement, and no outer optimization over the compatible CoPs is needed (compare Proposition 7.1). Under $\mathcal{M}_T$, taking $o = c^\star$ gives back the bounds of Section 6.

Lemma 9.2 (Fixed-reference lower bounds for taxel measurements)**.** *Under Assumption 5.1, for every $q = (a, b)$ and $o \in \mathbb{R}^2$,*

$$\underline{h}^o_{\mathcal{M}_F}(q; y_F) = \mu \min_{\substack{c_i \in \overline{P_i} \\ i=1,\dots,r}} \sum_{i=1}^{r} F_i \left\| a + bJ(c_i - o) \right\|_2, \tag{87}$$

and $\underline{h}^o_{\mathcal{M}_T}(q; y_T)$ is the same minimum with the additional constraint $\sum_{i=1}^r F_i c_i = f_N c$.

Proof. ($\geq$) For a compatible ν_N, its local CoPs are feasible (Lemma 5.5(i)–(ii)), and Lemma 5.5(iii) with $c = o$ bounds $\int_P g(x; q, o) \, d\nu_N$ from below by the objective. ($\leq$) For feasible c_i, $\sum_i F_i \delta_{c_i}$ is compatible and attains the objective (converse part of Lemma 5.5). The minimum exists, since the feasible set is compact and nonempty ($c \in \mathcal{C}_{\mathcal{M}_F}(y_F)$ for $\mathcal{M}_T$) and the objective continuous. □

Definition 9.3 (Load factor)**.** *For a nonzero load $w = (f, \tau) \in \mathbb{R}^2 \times \mathbb{R}$, expressed about o, the load factor of ν_N is $s^\star_o(\nu_N; w) \triangleq \max\{s \geq 0 : sw \in \mathcal{W}_o(\nu_N)\}$, and the certified load factor is*

$$\underline{s}_{\mathcal{M}}(w; y) \triangleq \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} s^\star_o(\nu_N; w). \tag{88}$$

The load w is carried by every compatible measure if and only if $\underline{s}_{\mathcal{M}}(w; y) \geq 1$. More generally, a contact whose normal force is scaled by λ carries w if $\lambda \, \underline{s}_{\mathcal{M}}(w; y) \geq 1$, since $\mathcal{W}_o(\lambda \nu_N) = \lambda \mathcal{W}_o(\nu_N)$.

Proposition 9.4 (Certified load factor)**.** *Let $w = (f, \tau) \neq 0$ and $o \in \mathbb{R}^2$.*

(i) For every measurement operator,

$$\boxed{\underline{s}_{\mathcal{M}}(w;y) = \max\left\{s \geq 0 :\ sw \in \mathcal{W}_o^-(y)\right\} = \inf_{q:\,\langle q,w\rangle=1} \underline{h}^o_{\mathcal{M}}(q;y).} \tag{89}$$

(ii) Under Assumption 5.1 with polygonal closures $\overline{P_i} = \{x : A_i x \leq d_i\}$, for $\sigma \in \{-1,+1\}$ let

$$\boxed{\begin{aligned} V_\sigma \triangleq \mu \inf_{a\in\mathbb{R}^2,\, b\in\mathbb{R},\, u_i\in\mathbb{R}^2} \quad & \sum_{i=1}^{r} F_i \left\| a + J(u_i - b\,o)\right\|_2 \\ \text{subject to} \quad & a^\top f + b\tau = 1, \qquad \sigma b \geq 0, \\ & \sigma\left(A_i u_i - b\,d_i\right) \leq 0, \qquad i = 1,\dots,r, \\ & \textstyle\sum_{i=1}^r F_i u_i = b\, f_N c \qquad (\mathcal{M}_T \text{ only}), \end{aligned}} \tag{90}$$

with $V_\sigma = +\infty$ if infeasible. Then $\underline{s}_{\mathcal{M}}(w;y) = \min\{V_{+1}, V_{-1}\}$ for $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$, and each V_σ is an SOCP.

(iii) If $(a^\star, b^\star, u^\star)$ is optimal for the smaller of the two programs, let $c_i = u_i^\star / b^\star$ if $b^\star \neq 0$, and let (c_i) be any feasible point of Lemma 9.2 otherwise. Then $\hat{\nu}_N = \sum_i F_i \delta_{c_i}$ is compatible and $s_o^\star(\hat{\nu}_N; w) = \underline{s}_{\mathcal{M}}(w;y)$: the certificate is attained by a compatible measure.

(iv) (Several contacts.) Let K contacts act on the same rigid object, with independent measurements y_k of their normal force measures ν_N^k, the same μ, and patches identified with subsets of a common plane (common coordinates x, torques about the common normal). Assume that the wrench components out of this plane are balanced without friction. The load w is then carried if $w \in \mathcal{W}_o(\nu_N^1) \oplus \cdots \oplus \mathcal{W}_o(\nu_N^K)$ (Minkowski sum). The certified load factor of the grasp, the infimum over all compatible $(\nu_N^1, \dots, \nu_N^K)$ of $\max\{s \geq 0 : sw \in \bigoplus_k \mathcal{W}_o(\nu_N^k)\}$, is

$$\boxed{\underline{s}_{\text{grasp}}(w) = \inf_{q:\,\langle q,w\rangle=1} \sum_{k=1}^{K} \underline{h}^o_{\mathcal{M}^k}(q;y_k),} \tag{91}$$

and (ii) holds with the objectives of the K contacts summed, a shared (a,b), and separate variables u_i^k. Since $\bigoplus_k \mathcal{W}_o^-(y_k) \subseteq \bigcap_{(\nu_N^k)} \bigoplus_k \mathcal{W}_o(\nu_N^k)$, this is at least the load factor of the sum of the guaranteed sets of the individual contacts.

Proof sketch. (i) Fix ν_N. Since $\mathcal{W}_o(\nu_N)$ is convex, compact and contains 0, $\{s \geq 0 : sw \in \mathcal{W}_o(\nu_N)\} = [0, s_o^\star(\nu_N; w)]$ with $s_o^\star < \infty$. Hence $sw \in \mathcal{W}_o^-(y)$ if and only if $s \leq s_o^\star(\nu_N; w)$ for all compatible ν_N, which gives the first equality. A closed convex set is the set of points satisfying $\langle q, \cdot\rangle \leq h(q)$ for all q [4, Section 13]. For $\langle q, w\rangle \leq 0$ this holds for every $s \geq 0$, since $h_{\mathcal{W}_o(\nu_N)} \geq 0$. Therefore $s_o^\star(\nu_N; w) = \inf_{\langle q,w\rangle>0} h_{\mathcal{W}_o(\nu_N)}(q)/\langle q, w\rangle = \inf_{\langle q,w\rangle=1} h_{\mathcal{W}_o(\nu_N)}(q)$ by positive homogeneity (the constraint set is nonempty since $w \neq 0$). Exchanging the infima over ν_N and over q gives the second equality.

(ii) By (i) and Lemma 9.2, $\underline{s}_{\mathcal{M}}(w;y)$ is the infimum of $\mu \sum_i F_i \|a + bJ(c_i - o)\|_2$ jointly over (a,b) with $a^\top f + b\tau = 1$ and over the feasible c_i. The objective is bilinear in (b, c_i). Split the feasible set by the sign of b and substitute $u_i = bc_i$. For $\sigma b > 0$, this is a bijection between the feasible c_i and the u_i with $u_i/b \in \overline{P_i}$, i.e., $\sigma(A_i u_i - b\,d_i) \leq 0$, and $\sum_i F_i u_i = b\, f_N c$. Moreover $bJ(c_i - o) = J(u_i - b\,o)$, so the objective is preserved. For $b = 0$, the original objective is $\mu f_N \|a\|_2$ for every feasible (c_i), and feasible points exist. In the new variables, $\sigma A_i u_i \leq 0$ forces $u_i = 0$ because $\overline{P_i}$ is bounded, and the objective is again $\mu f_N \|a\|_2$. The objective is a sum of norms of affine functions and the constraints are linear, so epigraph variables give an SOCP.

(iii) $\hat{\nu}_N$ is compatible (converse part of Lemma 5.5). With $q^\star = (a^\star, b^\star)$, by (i), $s_o^\star(\hat{\nu}_N; w) \leq h_{\mathcal{W}_o(\hat{\nu}_N)}(q^\star) = \mu \sum_i F_i \|a^\star + b^\star J(c_i - o)\|_2 = \underline{s}_{\mathcal{M}}(w;y) \leq s_o^\star(\hat{\nu}_N; w)$.

(iv) For fixed measures, $\bigoplus_k \mathcal{W}_o(\nu_N^k)$ is convex, compact, contains 0, and has support function $\sum_k h_{\mathcal{W}_o(\nu_N^k)}$. The argument of (i) applies. The compatible family is a product of the individual families, and the k-th term of the sum depends only on ν_N^k, so for fixed q the infimum of the sum is the sum of the infima. For the inclusion, if $w_k \in \mathcal{W}_o^-(y_k)$ for each k, then $w_k \in \mathcal{W}_o(\nu_N^k)$ for every compatible ν_N^k, so $\sum_k w_k \in \bigoplus_k \mathcal{W}_o(\nu_N^k)$. □

Remark 9.5 (On Proposition 9.4)**.**

- *Why not a grid of directions. $\underline{h}^o_{\mathcal{M}}(\cdot; y)$ is an infimum of convex functions of q, and is not convex in general. Minimizing it over a finite set of directions gives a value $\geq \underline{s}_{\mathcal{M}}(w; y)$, which may certify a load that is not guaranteed. Minimizing jointly over q and the c_i, after the change of variables $u_i = bc_i$, gives the exact value as a convex program.*

- *Coarse case. With a single taxel and $o = c$ ($\mathcal{M}_0$), the constraint forces $c_1 = c$, and (ii) gives $\underline{s} = \mu f_N / \|f\|_2$ if $\tau = 0$ and $\underline{s} = 0$ if $\tau \neq 0$ (take $a = 0$, $b = 1/\tau$). The coarse measurement never certifies a torsional load (Theorem 4.3).*

- *Taxel boundaries. The value is exact for the compatible family (46), defined with the taxel closures. For the stricter preimage $\mathcal{M}_F^{-1}(y_F)$ it is a valid, possibly conservative, certificate (Lemma 3.12).*

- *Scope. The certificate is quasi-static and applies to the measured state. In a feedback loop the load and the contact change between decisions, which requires a separate margin. In (iv), the assumption on the out-of-plane components matters: with a free object, offset normal forces at opposing contacts create a couple about an in-plane axis that must be balanced by friction, which the planar model does not account for.*

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